Insiyab



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INTRODUCTION

Urban congestion is a major challenge in modern cities, especially with the rapid expansion of public transport systems like Riyadh Metro. Anticipating congestion levels helps improve commuter experience and operational efficiency.

In this project, we present ***Insiyab*** – an AI-based solution that combines:

- A trained LSTM model to predict congestion based on time, location, and route data.
- A pre-trained NLP model that understands user queries and converts them into structured inputs.

Insiyab provides intelligent, real-time predictions through a user-friendly interface, supporting the vision of smart and data-driven urban mobility.





OBJECTIVES

Predict Congestion Levels

Develop a deep learning model to accurately forecast metro congestion based on spatial and temporal features.

Enable Natural Interaction

Use a pre-trained NLP model to interpret user-described trips and convert them into structured input for prediction.

Promote Smart Mobility

Support smarter transportation systems through AI-driven solutions aligned with sustainable urban development goals.



METHODOLOGY

- **A pretrained NLP** model to understand user input
- **A custom LSTM** model we trained to predict metro congestion

Goal:

- Let users ask naturally (e.g. “Is Al Batha crowded on Monday at 8 AM?”) and get smart, instant responses
- Behind the scenes: data generation, preprocessing, and model integration



METHODOLOGY

Data Collection & Features

- No public Riyadh Metro data was available → we created a synthetic structured dataset (10,000 rows)
- Simulates realistic metro trips and congestion

Data Preprocessing

- Cleaning: Removed missing/incomplete data
- Encoding: Transformed categorical columns into numbers
- Scaling: Normalized numeric fields (e.g. Hour)
- Balancing: Applied resampling to fix label imbalance (Medium was dominant)



METHODOLOGY

Pretrained NLP Model

- Used facebook/bart-large-mnli
- Understands user questions and extracts station, day, and time

Custom LSTM Model

- Trained on the processed dataset
- Predicts congestion level: Low / Medium / High

Model Integration

- The NLP model extracts structured features from user queries
- These features are passed to the LSTM model
- Result is displayed to the user via a Gradio web interface
- Simple, responsive, and human-friendly interaction



PERFORMANCE ANALYSIS

- The model achieved an overall accuracy of **95%**, with balanced precision, recall, and F1–scores across classes.
- **For class 0**, it showed **94%** precision and **100%** recall, indicating no false negatives.
- **For class 1**, precision was **92%** and recall was **100%**, also with no false negatives.
- **For class 2**, the model had **100%** precision but only **84%** recall, reflecting some false negatives.
- The confusion matrix shows that class 2 had the most misclassifications (3 misclassified as 0 and 4 as 1), while classes 0 and 1 were classified perfectly.
- Overall, the model demonstrates strong performance, especially in precision, but recall for class 2 could be improved.



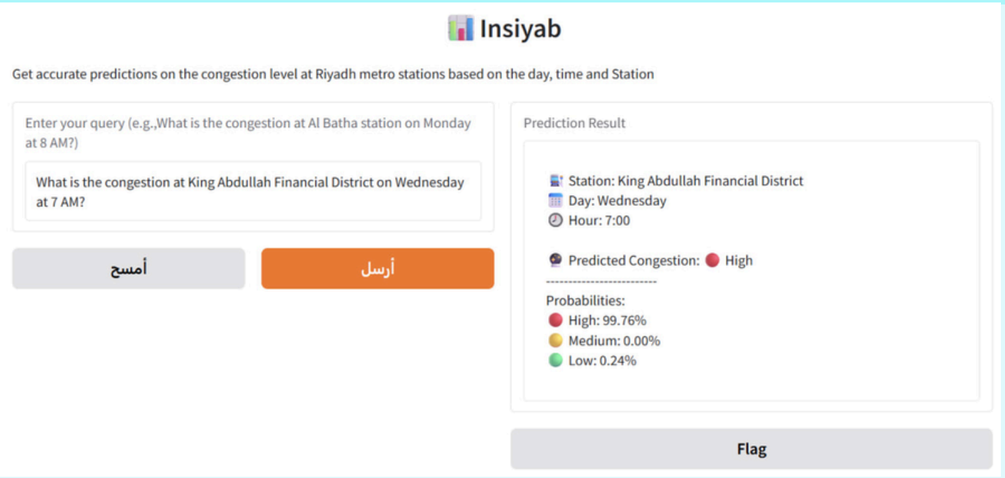
PERFORMANCE ANALYSIS

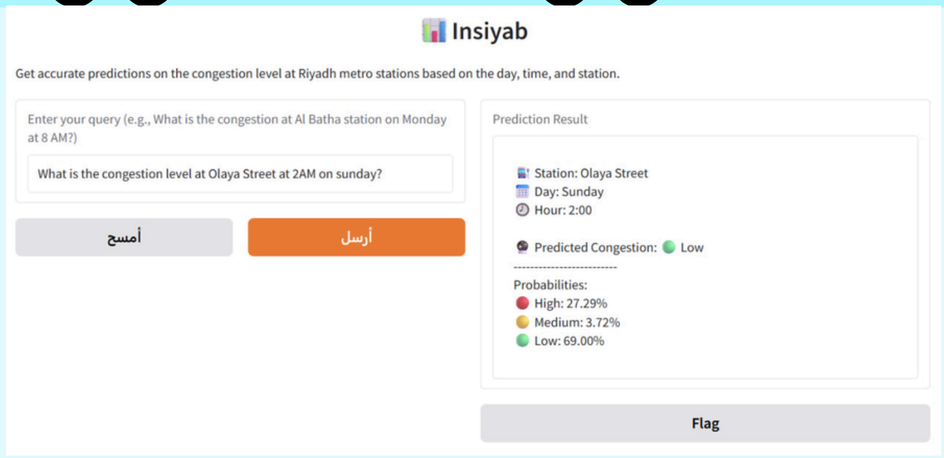
Class	Precision	Recall	F1-Score
0	94%	100%	97%
1	92%	100%	96%
2	100%	84%	91%

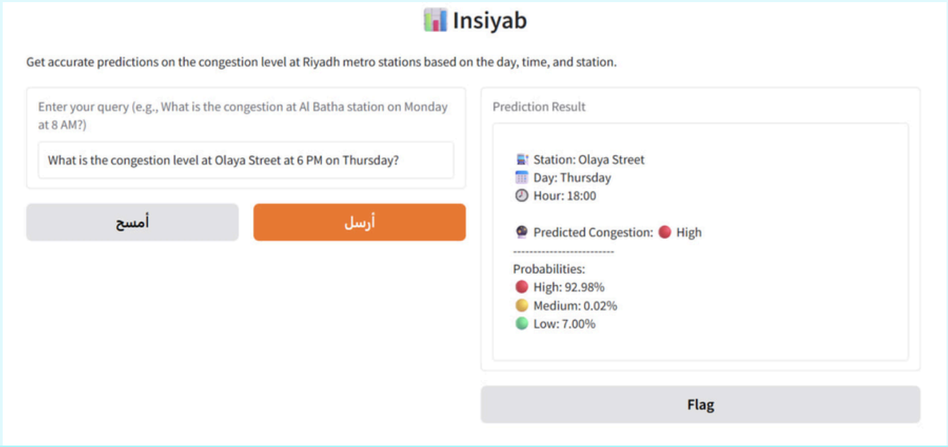


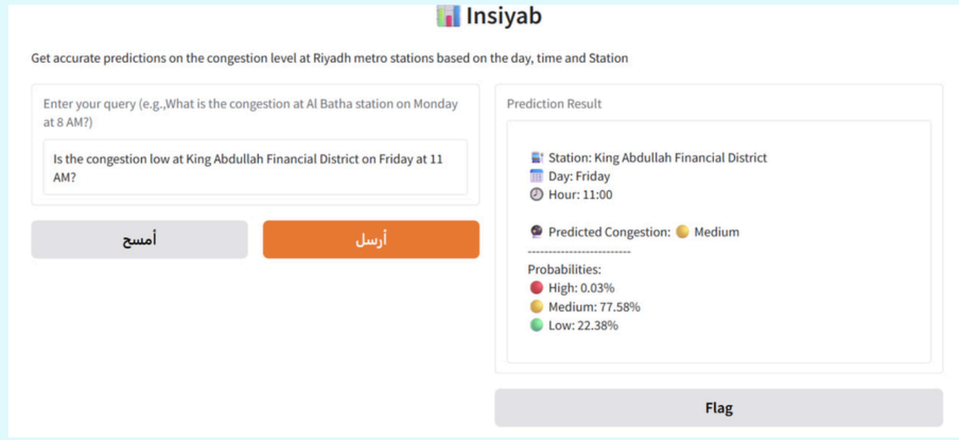
RESULTS AND DISCUSSION











LIMITATIONS

1. **Reliance on Synthetic Data:**
   The model was trained on synthetic data due to the absence of real data, which may not fully capture actual passenger behavior.

2. **Lack of Advanced Context Integration:**
   Factors such as major events, weather conditions, or schedule changes, which significantly impact congestion, have not been integrated into the system.

3. **Integration with Other Transport Systems:**
   Future integration with other transportation modes, such as smart buses, faces challenges related to infrastructure and data format standardization.



CONCLUSION

The **Insiyab** project demonstrates the application of artificial intelligence in addressing congestion challenges at Riyadh Metro.With an accuracy of up to 95%, the system effectively predicts congestion levels .
In the future, the focus will be on enhancing the user interface to better align with real-world standards. Real-time data integration will be implemented to improve prediction accuracy, along with the addition of technologies like speech recognition and interactive guidance. Additionally, future integration with other transportation modes, such as smart buses, will enhance the system's flexibility and increase its utility.
Insiyab is a scalable solution that aligns with Saudi Arabia's Vision 2030 and contributes to the development of smart transportation infrastructure.




“LET'S START USING PUBLIC TRANSPORTATION FOR A SUSTAINABLE FUTURE.”

THANK YOU

